8TH GRADE MATH • SUPPORT NOTES

Unit 2: Exponents & Scientific Notation

NC.8.EE.1, NC.8.EE.3, NC.8.EE.4

Name: ______________________ Date: ______________________

Class: ______________________ Period: ______________________

What Is In This Packet

- **Warm-Up:** what exponents mean + powers of 10.
- **Guided Notes 2.1–2.6:** exponent rules & scientific notation.
- **Practice** and **Answer Key**.

The One Big Idea of This Unit

An exponent is just a **shortcut for multiplying the same number** over and over.

★ Warm-Up: Skills We Need

Multiplying by 10s (place value)

Power	Means	Value
10^1	10	______
10^2	10×10	______
10^3	$10\times10\times10$	______

The exponent = how many zeros! 10^5 = 1 with 5 zeros = 100,000.

Words to Know

Base — the number being multiplied

Exponent / Power — how many times you multiply the base

Coefficient — the number in front (in 4×10^3, the 4)

Reciprocal — the flip of a number: reciprocal of 5 is 1/5

★ Expanded vs. Short

$$2^3 = 2\times2\times2 = 8$$

base

exponent

2.1 Introduction to Exponents

NC.8.EE.1

Expand & Condense

Short form	Expanded	Value
3^4	$3\times3\times3\times3$	______
5^2	______	______
2^5	______	______

Parentheses Change Everything

$(-2)^2 = (-2)(-2) = \mathbf{+4}$

$-2^2 = -(2\times2) = \mathbf{-4}$

Tip: a **negative** base with an **even** exponent → positive. With an **odd** exponent → negative.

2.2 Multiply & Divide Powers

NC.8.EE.1

Same Base Rules

Multiply → ADD the exponents. $x^a \times x^b = x^{a+b}$

Divide → SUBTRACT the exponents. $x^a \div x^b = x^{a-b}$

This only works when the **base is the same**.

I Do

$2^3 \times 2^4 = 2^{3+4} = \mathbf{2^7}$

$x^5 \div x^2 = x^{5-2} = \mathbf{x^3}$

You Try

$5^2 \times 5^5 = 5$______ $a^7 \div a^3 = a$______

2.3 Power-to-a-Power, Zero & Negative Exponents

NC.8.EE.1

Three More Rules

Power to a power → MULTIPLY. $(x^3)^4 = x^{3\times4} = x^{12}$

Zero exponent → always **1**. $7^0 = 1$

Negative exponent → flip it (reciprocal). $2^{-3} = 1/2^3 = 1/8$

I Do

$(3^2)^3 = 3^{2\times3} = \mathbf{3^6}$

$10^{-2} = 1/10^2 =$ **1/100**

You Try

$(x^4)^2 = x$______ $9^0 =$ ______ $5^{-1} =$ ______

2.4 Scientific Notation

NC.8.EE.3

The Form

$$a \times 10^n \quad (\text{where } 1 \le a < 10)$$

Big number → **positive** exponent. **Small** number (less than 1) → **negative** exponent.

How to Convert

Direction	Given	Answer	How
To Sci	45,000,000	______	move decimal 7 left
To Sci	0.00032	______	move decimal 4 right
To Standard	6.02×10^8	______	move 8 right

Watch Out

The front number must be between 1 and 10. 45×10^3 is **not** done — fix it to 4.5×10^4.

2.5 Operations with Scientific Notation

NC.8.EE.4

Multiply or Divide

1. Multiply (or divide) the **front numbers**.
2. Add (multiply) or subtract (divide) the **exponents**.
3. Fix the front number to be between 1 and 10 if needed.

I Do

$(3\times10^4)(2\times10^5) = (3\times2)\times10^{4+5} = \mathbf{6\times10^9}$

$(8\times10^7) \div (4\times10^3) = (8\div4)\times10^{7-3} = \mathbf{2\times10^4}$

You Try

$(2\times10^3)(4\times10^2) =$ ______ $\times\ 10$______

2.6 Word Problems

NC.8.EE.4

Plan Every Word Problem the Same Way

1. Underline the two numbers.
2. Decide the operation (per second? → divide. total? → multiply).
3. Set up in scientific notation.
4. Solve and fix the front number.

I Do

The Sun is 9.3×10^7 miles away. Light goes 1.86×10^5 miles/sec. How long to reach Earth?

$(9.3\times10^7) \div (1.86\times10^5) =$ **5×10^2** $= 500$ seconds.

Part 2 Practice

Simplify (write with one base)

#	Problem	Answer
1	$4^3 \times 4^2$	
2	$x^7 \div x^4$	
3	$(2^3)^2$	
4	5^0	
5	3^{-2}	
6	$(y^4)^3 \times y$	

Scientific Notation

1. Write 0.00056 in scientific notation: ______________

2. Write 7.2×10^6 in standard form: ______________

3. $(5 \times 10^3)(3 \times 10^4)$ = ______________

★ ANSWER KEY ★

Guided Notes

Warm-up: 10,100,1000. **2.1:** $3^4=81$; $5^2=5\times5=25$; $2^5=2\times2\times2\times2\times2=32$.

2.2: $5^2\times5^5=5^7$; $a^7\div a^3=a^4$. **2.3:** $(x^4)^2=x^8$; $9^0=1$; $5^{-1}=1/5$.

2.4: 4.5×10^7; 3.2×10^{-4}; 602,000,000. **2.5:** 8×10^5.

Practice

1) 4^5 2) x^3 3) 2^6 4) 1 5) 1/9 6) y^{13}

Sci: 1) 5.6×10^{-4} 2) 7,200,000 3) 1.5×10^8

End of Unit 2 support packet.